UCS503- Software Engineering Lab

(A typical Specimen of Cover Page & Title Page)

DREAMS2REALITY: A MINIMALIST TWO-ELECTRODE NON-INVASIVE BCI FOR REAL-TIME SLEEP-STAGE AND DREAM-STATE NEURAL DECODING

UCS 503 Software Engineering Project Report

Mid-Semester Evaluation

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UCS503- Software Engineering Lab

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1. Project Selection Phase

1.1 Software Bid

Executive Summary

The **Dreams2Reality** project delivers an intelligent, non-invasive software platform engineered to decode nocturnal neurophysiological dynamics, detect sleep-stage architecture, and classify dream-state cognitive mentation in real time. While conventional clinical polysomnography (PSG) setups require between 32 and 128 wet-electrode channels, prolonged 45-minute preparation, and capital expenditure exceeding $15,000, our software architecture is optimized for an ultra-lightweight, two-electrode dry temporal montage ($T_7\text{--}T_8$). By leveraging statistical Analysis of Variance (ANOVA F-score) channel selection and Riemannian geometry via Euclidean Covariance Alignment (EA), Dreams2Reality achieves high-fidelity dream-state feature separation while eliminating inter-subject skull impedance variability without requiring cumbersome daily recalibration.

Problem Statement & Market Opportunity

Over 30% of the adult population suffers from chronic sleep disturbances, nocturnal trauma intrusions, and parasomnias (such as PTSD-induced recurring nightmares, night terrors, and REM sleep behavior disorders). In psychiatric diagnostics and nocturnal sleep medicine, clinicians face an acute handicap: **the absence of objective, real-time quantification of nocturnal dream experiences.** Currently, clinicians rely almost exclusively on subjective retrospective self-reports obtained from patients upon awakening. These reports suffer from extreme recall decay, emotional bias, or complete amnesia.

While clinical polysomnography records physiological signals overnight, the physical encumbrance of dozens of scalp wires and adhesive paste induces the well-documented *"first-night effect,"* wherein the patient's sleep architecture is severely fragmented by physical discomfort, obscuring natural rapid eye movement (REM) episodes. There is an urgent clinical need for an unobtrusive, software-centric monitoring platform that continuously streams raw electroencephalography (EEG) from an ultra-minimalist dual-sensor headband, translates spontaneous neural fluctuations into structured dream-state semantic categories (motor agitation, threat/trauma intrusion, visual immersion, conversational mentation), and generates an automated clinical morning summary.

Technical Feasibility & Innovation

- **Electrode Dimensionality Reduction:** Clinical studies demonstrate that Rapid Eye Movement (REM) dream states and sensorimotor motor hallucinations produce distinct localized power shifts in the bilateral temporal and fronto-temporal regions. By isolating channels T_7 and T_8, our pipeline captures temporal beta/gamma desynchronization and asymmetric theta dynamics without requiring whole-scalp coverage.

- **Cross-Subject Domain Adaptation:** The principal obstacle in non-invasive EEG software is inter-subject variability arising from differences in cranial bone thickness, hair density, and electrode contact impedance. Dreams2Reality addresses this by projecting spatial covariance matrices into a standardized Euclidean reference space: $$\tilde{C}_i = R^{-\frac{1}{2}} C_i R^{-\frac{1}{2}}$$ where R represents the reference Riemannian geometric mean covariance of the calibration baseline, enabling zero-shot transfer across distinct patients.
- **Software-Centric Simulation Engine:** For the UCS503 engineering evaluation, physical hardware dependency is abstracted via an asynchronous telemetry playback server. The engine ingests clinical polysomnographic recordings from the PhysioNet Sleep-EDFx benchmark database, streaming 128 Hz packetized data over WebSockets to mimic an active bedside device.

Economic and Operational Feasibility

The software utilizes open-source programming stacks (Python, PyTorch, SciPy, MNE, FastAPI, React/WebSockets), resulting in zero software licensing expenditure. The inference engine requires fewer than 0.65 MFLOPs per 2.0-second sliding window, allowing execution on modest edge microprocessors (such as Raspberry Pi 4/5 or local bedside PC) with an average latency of < 15 ms, ensuring zero cloud-dependency and strict patient data confidentiality.

1.2 Project Overview

System Concept

Dreams2Reality is an intelligent, real-time biomedical software pipeline that bridges neurophysiological data acquisition and clinical cognitive assessment. The software continuously processes dual-channel EEG data, detects sleep-stage hypnograms (Wake, N1, N2, N3, REM), extracts high-dimensional non-linear features, and decodes the latent emotional valence, arousal, and semantic archetype of ongoing dream states.

```
┌──────────────────────────────────────────────┐
│        PhysioNet Sleep-EDFx Stream           │
│      (Dual-Channel T7-T8 @ 128 Hz)           │
└──────────────────────┬───────────────────────┘
                       │ Asynchronous WebSocket
                       ▼
┌──────────────────────────────────────────────┐
│        FastAPI Signal Preprocessor           │
│  [0.5-45 Hz Bandpass] + [50 Hz Notch Filter]│
└──────────────────────┬───────────────────────┘
                       │ 2.0s Windows / 0.5s Stride
                       ▼
┌──────────────────────────────────────────────┐
│         Feature Extraction Engine            │
│     • Differential Entropy (DE)              │
│     • DASM / RASM Temporal Asymmetry         │
│     • Euclidean Covariance Alignment (EA)    │
└──────────────────────┬───────────────────────┘
                       │ Normalized Feature Vector
                       ▼
┌──────────────────────────────────────────────┐
│          DC-CTA Neural Classifier            │
│ (Dual-Channel Cross-Temporal Attention Net)  │
└──────────────────────┬───────────────────────┘
                       │ Decoded Latent Vector
                       ▼
┌──────────────────────────────────────────────┐
│      Generative Narrative & Alert Engine     │
│   • Hall-Van de Castle Semantic Classifier   │
│   • Real-Time Bedside Nightmare Alerts       │
│   • Longitudinal Clinical Sleep/Dream Log    │
└──────────────────────────────────────────────┘
```

Core Architectural Pillars

1. **Telemetry & Signal Acquisition Subsystem:** Simulates real-time digital EEG telemetry using an asynchronous playback daemon reading calibrated Sleep-EDFx EDF records. It applies 4th-

order zero-phase Butterworth filtering ($0.5\text{--}45\text{ Hz}$) and adaptive baseline drift suppression.

2. **Feature Engineering & Dimensionality Subsystem:** Calculates Differential Entropy (DE) across Theta ($4\text{--}8\text{ Hz}$), Alpha ($8\text{--}12\text{ Hz}$), Beta ($12\text{--}30\text{ Hz}$), and Gamma ($30\text{--}45\text{ Hz}$) sub-bands. Computes Differential Asymmetry (DASM) and Rational Asymmetry (RASM) between temporal channels (T_7 and T_8) to monitor affective valence.
3. **Machine Learning Inference Subsystem:** Employs a lightweight Dual-Channel Cross-Temporal Attention Network (DC-CTA Net) that maps temporal dynamics into five standardized dream mentation archetypes derived from the Hall-Van de Castle clinical coding system.
4. **Presentation & Clinical Web Dashboard Subsystem:** A responsive web portal providing real-time hypnogram rendering, raw wave inspection, live trauma/nightmare spike alerts, and an automated Morning Clinical Dream Summary Report.

2. Analysis Phase

2.1 Use Cases

2.1.1 Use-Case Diagrams

The system models interactions between three distinct actors:

- **Sleeper / Patient (Data Source):** The monitored individual providing simulated or recorded real-time neuro-electrical telemetry.
- **Clinician / Sleep Therapist:** Medical professional reviewing overnight hypnograms, inspecting detected nightmare episodes, and evaluating patient sleep quality.
- **System Administrator:** Technical operator managing system calibration parameters, database backups, and simulated stream feeds.

```
                       DREAMS2REALITY SYSTEM BOUNDARY
+-------------------------------------------------------------------------+
|                                                                         |
|   (( UC-01: Ingest Telemetry Stream )) <-------- (Sleeper/Streamer)     |
|                 |                                                       |
|                 +--- <> ---> (( UC-02: Filter Noise & Drift )) |
|                                                    |                    |
|                                                    v                    |
|                                  (( UC-03: Align Riemannian Cov ))     |
|                                                    |                    |
|                                                    v                    |
|   (( UC-04: Classify Dream State )) <--------------+                    |
|                 |                                                       |
|                 +--- <> ----> (( UC-05: Trigger Nightmare Alert))
|                 |                                  |                    |
|                 v                                  |                    |
|   (( UC-06: Generate Morning Summary ))            v                    |
|                 ^                         [ Clinical Neurologist /      |
|                 |                           Sleep Therapist ]          |
|   (( UC-07: Review Hypnogram History )) -----------+                    |
|                                                                         |
|   (( UC-08: Calibrate Sensor Offsets )) <------- [ System Admin ]       |
|                                                                         |
+-------------------------------------------------------------------------+
```

2.1.2 Use Case Templates

Use Case ID: UC-01
Use Case Name: Stream & Ingest Dual-Channel EEG Telemetry
Primary Actor: Sleeper / Simulated Telemetry Daemon
Stakeholders & Interests: Patient desires zero physical intrusion; Clinician requires uninterrupted continuous signal logging.
Pre-conditions: Telemetry playback daemon initialized with valid Sleep-EDFx record; WebSocket port 8000 open.
Post-conditions: Raw 128 Hz packetized data placed into real-time buffer queue.
Trigger: Clinician clicks "Start Overnight Monitoring Session" on dashboard.
Main Success Scenario:

1. System initializes WebSocket endpoint `/ws/eeg-stream`.
2. Telemetry daemon establishes persistent connection and streams packets at 128 samples/second.
3. Buffer verifies packet sequence checksums and begins 2.0s sliding-window assembly (256 samples).

Alternative Flow: In case of socket disconnection, buffer raises a warning flag and triggers automated reconnection with exponential backoff (1s, 2s, 4s).

Use Case ID: UC-04
Use Case Name: Real-Time Dream-State & Semantic Category Classification
Primary Actor: System ML Inference Engine
Secondary Actor: Clinical Neurologist / Sleep Therapist
Pre-conditions: Ingestion buffer contains at least one calibrated 2.0s epoch free from motion saturation (> 250 μV).
Post-conditions: Sleep stage (Wake, N1, N2, N3, REM), valence score [-1.0, +1.0], arousal [0.0, 1.0], and Hall-Van de Castle semantic theme are persisted and broadcast.
Trigger: Preprocessor signals completion of Differential Entropy and Euclidean Covariance computations.
Main Success Scenario:

1. Inference engine extracts 16-D feature vector across Theta, Alpha, Beta, Gamma bands.
2. Euclidean aligner projects spatial covariance to reference manifold.
3. DC-CTA Net evaluates tensor input and computes posterior class probabilities.
4. Decoded state is saved to datastore and transmitted via WebSocket to the live interface.

Use Case ID: UC-05
Use Case Name: Automated Bedside Nightmare & Trauma Distress Alert
Primary Actor: System Alert Dispatcher | **Secondary Actor:** Bedside Attendant
Pre-conditions: Active session in progress; patient confirmed in REM sleep.
Post-conditions: Priority visual/acoustic alert generated on clinician console.
Trigger: Valence drops below -0.75 and arousal exceeds 0.85 concurrently with beta/gamma surges for > 10s.
Main Success Scenario:

1. System detects sustained negative valence spike during REM sleep.
2. Distress classifier confirms category `Threat_Simulation_Panic`.
3. Bedside dashboard triggers priority visual strobe and acoustic notification.
4. Exact 60-second EEG snippet is archived for retrospective clinical evaluation.

2.2 Activity Diagram and Swimlane Diagrams

The operational lifecycle of Dreams2Reality spans four concurrent execution partitions (swimlanes):

```
+--------------------+---------------------+--------------------+--------------------
|  Dataset Telemetry | Signal Preprocessing| Machine Learning   |  Clinical Web
|      Streamer      |       Engine        |   Brain Engine     |    Dashboard
+--------------------+---------------------+--------------------+--------------------
|         |          |          |          |         |          |         |
|         |          |          |          |         |          |  [Clinician clicks
|         |          |          |          |         |          |   Start Session]
|         |          |          |          |         |          |         |
|  [Receive Start    |<---------+----------+---------+----------+---------+
|   Command]         |          |          |         |          |         |
|         |          |          |          |         |          |         |
|  [Stream 128 Hz    |          |          |         |          |         |
|   EDF Packets]     |          |          |         |          |         |
|         +--------->|  [Apply 0.5-45 Hz   |         |          |         |
|         |          |   Bandpass Filter]  |         |          |         |
|         |          |          |          |         |          |         |
|         |          |  [Apply 50 Hz       |         |          |         |
|         |          |   Notch Filter]     |         |          |         |
|         |          |          |          |         |          |         |
|         |          |   |         |          |         |          |
|         |          |     /         \     |         |          |         |
|         |          | [YES]         [NO]  |         |          |         |
|         |          |   |             |   |         |          |         |
|         |          | [Discard]  [Assemble|         |          |         |
|         |          |  Packet]   2.0s Win]|         |          |         |
|         |          |                 |   |         |          |         |
|         |          |                 +-->| [Compute Welch PSD |         |
|         |          |                     |  & Diff Entropy]   |         |
|         |          |                     |         |          |         |
|         |          |                     | [Calculate DASM/   |         |
|         |          |                     |  RASM Asymmetry]   |         |
|         |          |                     |         |          |         |
|         |          |                     | [Project Euclidean |         |
|         |          |                     |  Covariance (EA)]  |         |
|         |          |                     |         |          |         |
|         |          |                     | [Execute DC-CTA    |         |
|         |          |                     |  Net Inference]    |         |
|         |          |                     |         |          |         |
|         |          |                     |  |         |          |
|         |          |                     |     /         \    |         |
|         |          |                     |  [YES]        [NO] |         |
|         |          |                     |    |            |  |         |
|         |          |                     | [Evaluate    [Log  |         |
|         |          |                     |  Valence &  Sleep  |         |
|         |          |                     |  HVdC Theme] Stage]|         |
|         |          |                     |    |            |  |         |
|         |          |                     |    +-----+ +-----+ |         |
|         |          |                     |         |  |       |         |
```

```
|         |          |                    |  [Dispatch Event   |         |
|         |          |                    |   Telemetry Packet]|         |
|         |          |                    |         |          |         |
|         |          |                    |         +--------->|  [Update Real-Time
|         |          |                    |                    |   Hypnogram UI]
|         |          |                    |                    |         |
|         |          |                    |                    |    |
|         |          |                    |                    |     /         \
|         |          |                    |                    |  [YES]        [NO]
|         |          |                    |                    |    |             |
|         |          |                    |                    | [Trigger   [Render
|         |          |                    |                    |  Acoustic   Normal
|         |          |                    |                    |  Alarm]     Wave]
|         |          |                    |                    |         |
|         |          |                    |                    |  [Session Ends:
|         |          |                    |                    |   Compile Morning
|         |          |                    |                    |   Summary PDF]
|         |          |                    |                    |         |
+---------+----------+--------------------+--------------------+---------+----------
```

2.3 Data Flow Diagrams (DFDs)

2.3.1 DFD Level 0 (Context Diagram)

```
                          +--------------------------------+
                          |  Sleep-EDFx Playback Streamer |
                          |   (Simulated 2-Ch Electrode)   |
                          +---------------+----------------+
                                          |
                                          | Raw Dual-Channel EEG Packets
                                          | (128 Hz, T7-T8, Microvolts)
                                          v
                              +-----------------------+
                              |         0.0           |
                              |     DREAMS2REALITY    |
                              |   CORE BCI PLATFORM   |
                              +-----------+-----------+
                                          |
                       +------------------+------------------+
                       |                                     |
                       | Real-Time Stream,                   | Morning Clinical
                       | Live Hypnogram & Alerts             | Dream Summary PDF
                       v                                     v
            +------------------------+            +------------------------+
            |  Clinical Neurologist |            | Medical Archival &     |
            |   / Sleep Attendant    |            | Electronic Health Rec |
            +------------------------+            +------------------------+
```

2.3.2 DFD Level 1 (Modular Functional Pipeline)

```
[Sleep-EDFx Streamer]
         |
         | Raw Packets (T7, T8 @ 128 Hz)
         v
  +---------------+
  |      1.0      |
  | Ingestion &   |======== Raw Buffer Queue ========> (D1: Raw Stream Buffer)
  | Buffering     |
  +-------+-------+
          |
          | Sequenced 128 Hz Stream
          v
  +---------------+
  |      2.0      |
  | Digital Pre-  |<======== Filter Coefficients ===== (D2: System Config DB)
  | processing    |
  +-------+-------+
          |
          | Clean 2.0s Windows (0.5-45 Hz)
```

```
         v
+---------------+
|     3.0       |
| Feature Eng. |<======== Reference Covariance === (D3: Calibration Manifold)
| & Alignment  |
+-------+------+
        |
        | 16-D Normalized Feature Vector
        v
+---------------+
|     4.0       |
| DC-CTA Net   |<======== Neural Weights ========= (D4: Model Weights DB)
| Inference    |
+-------+------+
        |
        | Decoded Dream Vector (Stage, Valence, Arousal, Theme)
        v
+---------------+
|     5.0       |
| Presentation |----------------------------------> [Clinical Web Dashboard]
| & Reporting  |----------------------------------> [Morning PDF Report]
+---------------+
```

2.3.3 DFD Level 2 (Subsystem 3.0: Feature Engineering & Spatial Alignment)

```
Clean 2.0-Second Epoch (256 samples x 2 channels)
                    |
                    +--------------------------------------+
                    |                                      |
                    v                                      v
          +---------------------+                +--------------
          |         3.1         |                |        3.2
          |  Welch PSD Spectral|                | Spatial Covar
          |     Estimator       |                |    Computatio
          +---------+-----------+                +---------+----
                    |                                      |
                    | Band Powers                          | 2x2
                    | (Theta, Alpha, Beta, Gamma)          v
                    v                                +--------------
          +---------------------+                |        3.3
          |         3.4         |                | Euclidean Ali
          | Differential        |                | R^(-1/2) C_i
          | Entropy Calculator |                +---------+----
          +---------+-----------+                          |
                    |                                      | Ali
                    | 8-D DE Vectors                       v
                    +--------------------------------------+
                    |
                    v
          +---------------------+
          |         3.5         |
          | Temporal Asymmetry |
          |   (DASM / RASM)     |
          +---------+-----------+
                    |
                    | Asymmetry Indices (T7 vs T8)
                    v
          +---------------------+
          |         3.6         |
          | Feature Assembly    |
          | & Vector Packing    |
          +---------+-----------+
                    |
                    v
 16-Dimensional Normalized Neural Feature Tensor
```

2.4 Software Requirement Specification (SRS) in IEEE Format

1. Introduction

1.1 Purpose: This document specifies the software requirements for **Dreams2Reality**, conforming to the **IEEE Std 830-1998** standard.
1.2 Scope: Non-invasive assistive software platform ingesting dual-channel temporal EEG ($T_7\text{--}T_8$) sampled at 128 Hz, filtering noise, extracting non-linear features, projecting covariance onto Riemannian manifolds, and classifying sleep/dream mentation in real time.
1.3 Definitions: BCI (Brain-Computer Interface), PSG (Polysomnography), REM (Rapid Eye Movement), DE (Differential Entropy), EA (Euclidean Alignment), HVdC (Hall-Van de Castle taxonomy).

2. Overall Description

2.1 Product Perspective: Client-server biomedical monitoring package with FastAPI backend and low-latency React canvas frontend.
2.2 General Constraints: Must execute on standard 4-core CPU without discrete GPU, with end-to-end latency < 15 ms per 2.0-second epoch.

3. Specific Requirements

3.1 Functional Requirements

Req ID	Requirement Name	Detailed Specification
FR-01	Signal Telemetry Ingestion	Ingest dual-channel EEG at 128 Hz over asynchronous WebSockets from playback daemon.
FR-02	Digital Filtering	Apply 4th-order zero-phase Butterworth bandpass ($0.5\text{--}45\text{ Hz}$) and 50 Hz IIR notch filter.
FR-03	Artifact Rejection	Automatically flag and discard sliding windows exceeding $\pm 250\ \mu\text{V}$ peak-to-peak amplitude.
FR-04	Sliding Windowing	Segment stream into 2.0s epochs (256 samples) with 0.5s step stride (75% overlap).
FR-05	Differential Entropy	Compute Welch PSD and DE across Theta, Alpha, Beta, and Gamma frequency sub-bands.
FR-06	Hemispheric Asymmetry	Calculate DASM ($DE_{T7}-DE_{T8}$) and RASM (DE_{T7}/DE_{T8}) affective valence indices.

Req ID	Requirement Name	Detailed Specification
FR-07	Euclidean Alignment	Normalize spatial covariance matrices via reference Riemannian geometric mean to eliminate skull impedance drift.
FR-08	Dream Classification	Classify sleep stages and map REM mentation into Hall-Van de Castle semantic archetypes.
FR-09	Nightmare Distress Alarm	Trigger auditory/visual alert when valence < -0.75 and arousal > 0.85 during REM for > 10s.
FR-10	Morning PDF Export	Generate downloadable clinical summary with hypnogram, dream duration, and trauma timestamps.

3.2 Non-Functional Requirements

Req ID	Quality Attribute	Performance Metric & Benchmark
NFR-01	Inference Latency	End-to-end pipeline execution < 15 milliseconds per 2.0s epoch on standard edge CPU.
NFR-02	Batch Throughput	Sustain > 250 epochs/sec during historical dataset playback and retrospective scoring.
NFR-03	Reliability & Uptime	Maintain 99.5% uptime across continuous 10-hour nocturnal monitoring sessions.
NFR-04	Security & Privacy	Local encrypted storage (AES-256) and TLS-authenticated local WebSocket channels.
NFR-05	UI Usability & FPS	Maintain ≥ 30 FPS waveform rendering on HTML5 canvas with WCAG 2.1 AA contrast.
NFR-06	Modularity	Decoupled Python package architecture with > 85% automated unit test coverage.

2.5 User Stories and Story Cards

Story ID: US-01 | **Priority: Highest** | **Story Points: 5**
User Story: *As a Sleep Technologist, I want to stream 2-channel temporal EEG data at 128 Hz from simulated playback files, so that I can evaluate patients without needing physical 64-channel wet electrode caps.*
Acceptance Criteria:

- WebSocket connection establishes within 500 ms of session launch.
- Zero packet drop over 1-hour continuous playback stream at 128 Hz.
- Automated reconnection within 3 seconds of transient network drop.

Story ID: US-02 | **Priority: High** | **Story Points: 3**
User Story: *As a Biomedical Data Scientist, I want raw signals automatically filtered between 0.5 Hz and 45 Hz and notch-filtered at 50 Hz, so that powerline hum and baseline drift are purged before inference.*
Acceptance Criteria:

- Zero-phase distortion via forward-backward Butterworth IIR filter.
- Motion artifacts > ±250 µV flagged and excluded from downstream inference.
- Preprocessing execution latency under 3.0 ms per 2.0s epoch.

Story ID: US-03 | **Priority: High** | **Story Points: 8**
User Story: *As a Software Engineer, I want to project spatial covariance matrices into a standardized Euclidean space, so that the pre-trained neural network performs accurately on new patients without manual retraining.*
Acceptance Criteria:

- Calculates Riemannian reference mean covariance from initial 60-second baseline.
- Transforms covariance matrices via inverse matrix square roots.
- Maintains zero-shot cross-subject classification accuracy > 70% on unseen subjects.

Story ID: US-04 | **Priority: Critical** | **Story Points: 5**
User Story: *As a Bedside Caregiver, I want instant visual and acoustic warnings when a patient experiences intense nightmare distress, so that I can provide timely bedside comfort.*
Acceptance Criteria:

- Alert dispatches within 1.0s of detecting REM negative valence (< -0.75) and high arousal (> 0.85).
- Flashing amber visual strobe and gentle acoustic alarm on bedside terminal.
- Single-click acknowledgment and snooze logging attendant intervention time.

Story ID: US-05 | **Priority: Medium** | **Story Points: 5**
User Story: *As a Clinical Neurologist, I want an automated PDF report summarizing the night's sleep stages and dream categories, so that I can evaluate sleep architecture without manual scoring.*
Acceptance Criteria:

- Compiles multi-page clinical report within 5 seconds of session termination.
- Includes graphical hypnogram timeline, stage duration percentages, and trauma timestamps.
- Categorizes dream mentation into Hall-Van de Castle semantic themes.

--- END OF UCS 503 MID-SEMESTER EVALUATION REPORT ---